MILITARY SPECIFICATION

CELLULOSE NITRATE, PLASTIC SHEETS

1. SCOPE

1.1 This specification covers cellulose nitrate plastic sheets for propellant increment containers.

1.2 Classification.-The plastic sheets shall be of the following type:

Type I - with urea

Type II - with ethyl centralite

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-N-244 - Nitrocellulose
MIL-E-255 - Ethyl Centralite (Carbamite)
MIL-U-10866- UREA, Technical

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for for Inspection by Attributes.

PSC: 1376

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0 - 190

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite, "49 CFR 0-190 (latest revisions)".

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the National Classification Board, 1616 P. Street, N. W. Washington, D.C., 20036).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D638 - Tensile Properties of Plastics
D1898- Sampling of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

3. REQUIREMENTS

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3.1 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.2 Material

3.2.1 Composition.-The cellulose-nitrate sheets shall be a homogeneous mixture of the required ingredients. Ingredients used shall conform to the following requirements.

3.2.1.1 Nitrocellulose.-The nitrocellulose used for Type I composition shall have a nitrogen content of 11.00 ± 0.5 percent and meet the requirements of MIL-N-244. The nitrocellulose used for Type II composition shall be Grade D in accordance with MIL-N-244.

3.2.1.2 Camphor.-The camphor shall be of good commercial grade with a melting point range between 174.4° and 179.0°C. A 20 gram sample shall show no discoloration when heated in a glass test tube for 30 minutes in an oil bath at 150°C.

3.2.1.3 Urea.-The urea shall be in accordance with MIL-U-10866, Class I.

3.2.1.4 Ethyl Centralite.-The ethyl centralite shall be in accordance with MIL-E-255.

3.3 Color.-Cellulose nitrate plastic sheets shall be clear and transparent.

3.4 Dimensions.-The cellulose nitrate sheets shall conform to dimensions as specified by the contract or order.

3.5. Composition.-The cellulose nitrate plastic shall conform to the following compositions:

Type	I	II
	Percent	
Nitrocellulose	75.0 ± 2.0	75.0 ± 2.0
Camphor	25.0 ± 2.0	25.0 ± 2.0
Urea(Added)*	0.75 ± 0.1	
Ethyl Centralite (Added)*	-	0.75 ± 0.1

*Percent of the dry weight of nitrocellulose.

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3.6 Tensile strength and elongation.-The plastic sheets shall have a tensile strength of 5,000 lbs per square inch, minimum (min) and elongation of 10 percent (min) when tested as specified in 4.4.1.

3.7 Cure-time.-The plastic sheets shall be cured at ambient condition for two (2) weeks. The cure-time shall be calculated from the first day the plastic is molded.

3.8 Stability.-The plastic sheets shall not change the methyl violet paper to a salmon pink color in less than 40 minutes when tested in accordance with 4.4.2.

3.9 Workmanship.-The plastic shall be free from warpage, cracks, scratches, and other defects which may affect its useability and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contractor or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Specification number and date, together with identification and date of changes thereto.

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c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by the Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection (see 6.2)

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions fo 4.2.2. The first article sample, consisting of 10 sheets 1 by 1 1/2 ft (sufficient in size to perform the specified tests) shall be selected from consecutively produced sheets which have been manufactured using the same production processes, proc[illegible] [illegible] contract. All materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period of excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to signficantly affect product uniformity, as determined by the Government.

4.2.2 Inspections to be performed.-The finished sample shall be subjected by the Government to any or all of the examinations or tests specified in this specification.

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4.2.3 Rejection.-If the finished sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any component to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior to subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchange process, in accordance with the same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Inspection procedure

4.3.2.1 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with MIL-STD-105. Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each minor defect.

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Categories	Defects	Method of Inspection	Code No. (see 6.5)
Critical:	None defined.		
Major:	AQL 1.0 percent defect		
101.	Dimensions incorrect	Gage	01001
102.	Warpage	Visual	01002
103.	Cracks	Visual	01003
Minor:	AQL 2.5 percent defect		
201.	Scratches	Visual	01004
202.	Color	Visual	01005

4.4 Tests

4.4.1 Tensile strength and elongation - Code No. 02001.- The tensile strength and elongation test shall be in accordance with ASTM D638. Sampling for the test shall be conducted in accordance with ASTM D1898, Method C.

4.4.2 Stability - Code No. 03001.-The stability test shall be performed at 134.5°C in accordance with Method 404.1 of Standard MIL-STD-286.

4.4.3 Composition-Code No. 04001.-Each ingredient used in the composition shall be weighed and check weighed to conform with the requirements of 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging Level C.-Quantities of plastic sheets as specified (see 6.2) shall be packed to assure adequate protection against deterioration and damage from the supplier to the first receiving activity for immediate use.

5.2 Packing, Level C.-Plastic sheets packaged as specified in 5.1, shall be packed in uniform quantities in accordance with Department of Transportation (DOT) regulations, Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules as may be applicable to the mode of transporation.

5.3 Marking.-Unit and shipping containers shall be marked in accordance with MIL-STD-129 and DOT regulations. Shipping containers shall bear the DOT yellow "Flammable Solid" Pyroxylin Plastic label.

6. NOTES

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6.1 Intended use.-The plastic covered by this specification is intended for use in increment containers for propellant for the 60MM and 81MM Mortar.

6.2 Ordering data.-Procurement documents should specify the following:

a. the title, number and date of this specification.
b. Color when required (see 3.3)
c. The dimensions of the plastic required should be specified in the contract. (see 3.4)
d. Quantities of plastic per unit package required (see 5.1)
e. Provisions for submission of first article samples.

6.3 Warning.-The material covered by this specification is extremely flammable. Suitable precautions should be taken in machining specimens and in storage to reduce hazards involved. Because of the deteriorating effects of aging and of old embrittled materials in manufacture and to provide adequate product protection during shipment and storage.

6.4 Significant places.-For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29)

6.5 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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Army-MU

Preparing Activity:
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Project Number: 1376-A010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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